UCDAVIS VETERINARY MEDICINE

VET- 436 / 2026

Emergency and Critical Care

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Emergency presentations are very common in exotic mammals. Fortunately, **many concepts that you learn in domestic carnivores medicine are also applicable to small mammals.** Many presentations are the same such as seizures or dyspnea. Triaging, fluid therapy, CPR, and diagnostic management are very similar to dogs and cats. With that said, even if much can be generalized to small mammals, **there are still some important species-specific differences** to be aware of. This lecture will mainly highlight these differences and I will not go into very much details when it comes to general background knowledge such as fluid physiology, pathophysiology of shocks, and mechanisms of action of emergency drugs.

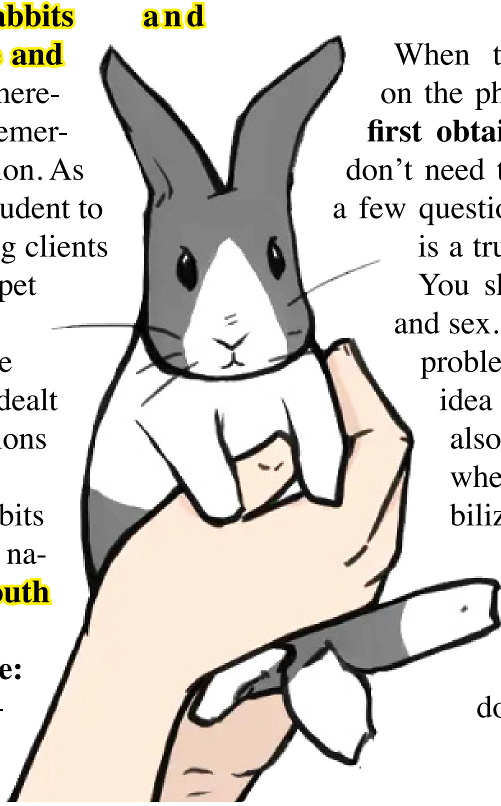
TRIAGE

General Concepts

Most of the small mammals that we see in practice are prey species (except the ferret). **Rabbits and rodents tend to hide signs of disease and have a high level of compensation.** Therefore, chronic illnesses often present as emergency because of sudden decompensation. As it can be hard to tell sometimes, it is prudent to err on the side of caution when advising clients whether or not they should bring their pet on emergency.

In any case, some presentations are more urgent than others and should be dealt with first. Life-threatening presentations mostly include the following:

- **Respiratory distress:** rabbits and rodents in particular are obligate nasal breathers, so if they are **open-mouth breathing,** it is always very serious.
- **Extreme lethargy or collapse:** this may be associated with shock, sepsis, extreme pain, or neurological diseases.
- **Seizures:** if animals are actively seizuring, they should be seen immediately and all stimuli should be reduced until they can be seen.
- **Severe trauma or uncontrolled hemorrhage:** there are immediate danger of shock, dehydration, sepsis, multi-organ failure or further deterioration.
- **Acute abdominal pain:** most underlying diseases are severe and many conditions causing acute abdominal pain lead to emergency surgeries.
- **Severe diarrhea or vomiting:** there are immediate and continuing risks of severe dehydration and electrolyte loss. In rabbits and rodents, this may raise the suspicion for dysbiosis or enterotoxemia, which should be addressed quickly. Rabbits and rodents seldom vomit, but ferrets can definitely vomit. Ferrets do not vomit frequently, so when they do, it tends to be specific to a gastric issue or a foreign body obstruction.



In Practice

When triaging small mammal patients on the phone or for a walk-in, you should **first obtain some basic information.** You don't need to take a full history, but just ask a few questions to have a sense of whether it is a true emergency and to get prepared. You should inquire about species, age, and sex. Go straight about asking what the problem is and its duration. Having an idea of the **estimated arrival time** is also useful to get ready and to decide whether the animal should be first stabilized by a closer veterinarian. When the animal arrives, you need to quickly evaluate its condition. You should first determine whether the animal is stable by doing a brief assessment that should include the following:

- Assess mentation: if the animal is comatose, you may have to start emergency treatment or CPR.

- Assess volemia and perfusion: you can have a quick idea of this by obtaining the **heart rate, palpating the femoral pulse, looking at the capillary refilling time (CRT), and obtaining a blood pressure measurement.** Animals in shock can either be tachycardic (compensated shock) or bradycardic (decompensated shock). CRT is a good estimate of peripheral perfusion. The femoral pulse can be hard to palpate in small mammals, but if you can feel it pretty well, then odds are that the blood pressure is adequate. **Indirect blood pressure measurements are not very accurate in small mammals,** especially when taken by oscillometry. Blood pressure also tends to be physiologically lower in rabbits than in other mammals (we typically use around 60-80 mmHg systolic as a cut-off).

- Assess hydration: hydration is not the same thing as volemia and it should be assessed separately. Skin tent and mucus membranes are used.

- Assess temperature: hypothermia is a sign of hypotension and is also a **strong negative prognosticator in rabbits and guinea pigs. Small mammals tend to lose heat rapidly** because of high surface area-to-volume ratio.

- Other stuff: you should also perform a quick cardiopulmonary auscultation to attempt to detect obvious issues such as a heart murmur, bradycardia, arrhythmias, and pulmonary noises. A quick abdominal palpation will allow to also feel obvious masses, abdominal distention or a large stomach in rabbits. Finally, a quick visual exam should be performed to look for wounds, masses, and other skin and musculoskeletal abnormalities.

When the animal is more stable, then you can finish collecting the history and complete the physical examination.

CARDIOPULMONARY RESUSCITATION

Getting Prepared

You obviously don't want to start looking for emergency drugs or try to decide what each team member should do when a patient is presented in cardiopulmonary arrest. So how do you get prepared?

First, you need to know and understand the different steps and current recommendations for CPR in veterinary medicine. These are nicely outlined by the RECOVER initiative. Make sure every member of your team is aware of these guidelines. You may provide visual aids in the main treatment area for easy reference. These should include a chart of drug volumes per weight for quick calculation. It may also be important to run **regular drills** so you can identify potential issues and improve your preparedness for CPR.

It is a good idea to **prepare emergency drugs in advance** when anesthetizing a patient so they can be injected quickly.

Another thing you need to have ready and well organized is your **crash cart**. It should be in the main treatment area and of easy access. It should contain the equipment in the following categories:

- Ventilation: various sizes endotracheal tubes, mainly for ferrets. Rabbits and rodents are hard to intubate and it is typically not feasible on emergency so alternatives should be present such as tight-fitting masks and supraglottic airway devices (V-gels).

- Vascular access: various sizes of intravenous catheters and also spinal needles and other needles for potential use as intraosseous catheters. A wound pack should be available in case a cut-down access is indicated.

- Emergency drugs: these include epinephrine, atropine, **glycopyrrolate (since many rabbits have atropinases, this should be available as well)**, bicarbonates, drug reversals (flumazenil, naloxone), lidocaine, regular and hypertonic saline, dextrose 50%, furosemide, and calcium gluconate.

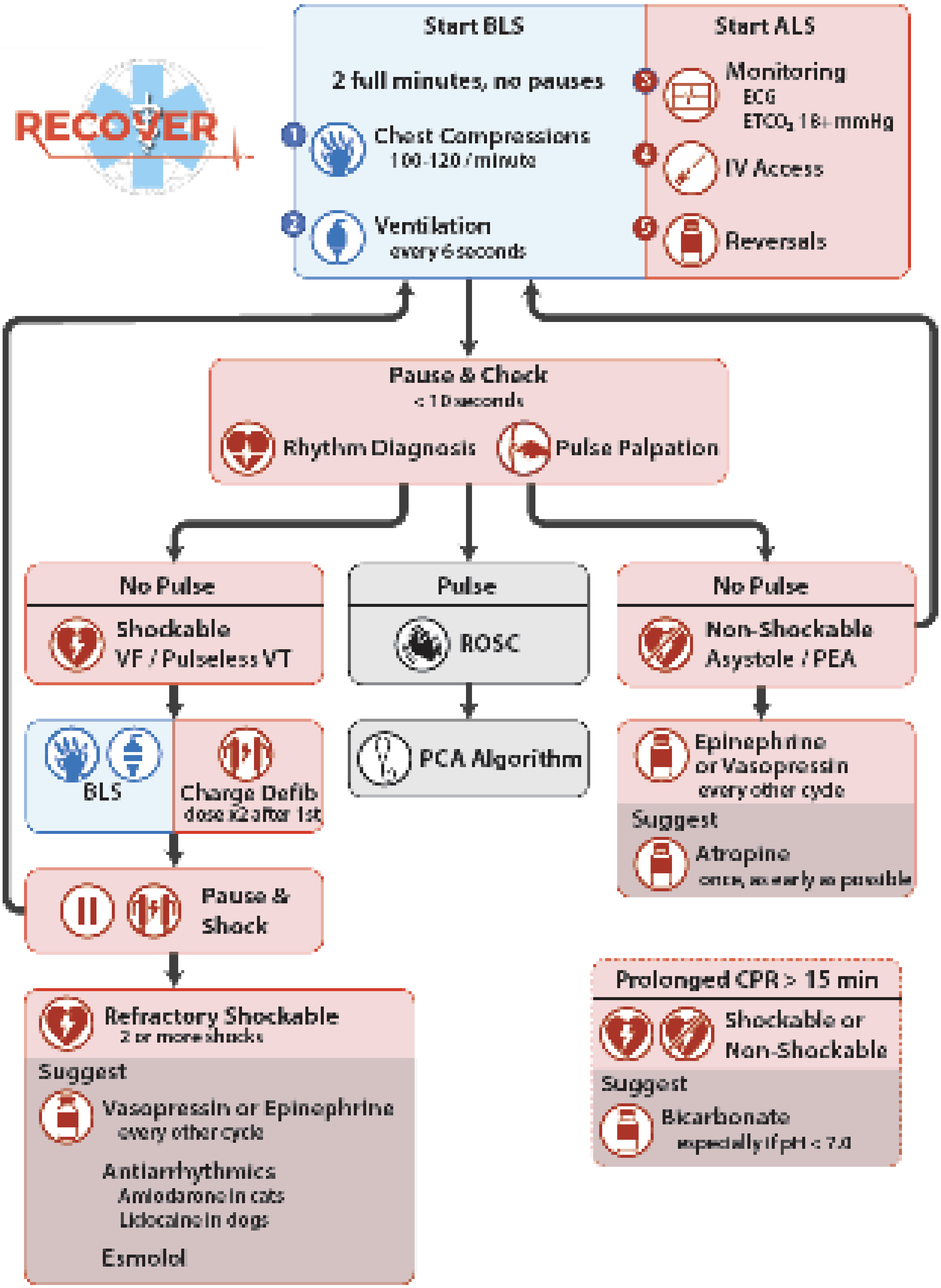
- Monitoring equipment: these can be on the main anesthetic machine of your treatment room and include a doppler unit, capnography, thermometer, pulse oximeter, ECG, and a stethoscope.

In Practice

The indications for CPR are similar as for other mammals and are basically an **unresponsive patient in cardiopulmonary arrest.** Small mammal species have a higher metabolic rate and increased oxygen demand when compared to domestic carnivores. In addition, many emergency procedures are more challenging to do such as placing catheters, collecting blood, and intubation. For these reasons, the **outcomes of CPR tend to be lower in small mammal species**.

See next page for the current RECOVER recommendations. Most, but not all, are applicable to small mammal species. As mentioned previously, intubation is nearly impossible to do on a crashing apneic rabbit or rodent. Because they are obligate nasal breathers,

CPR Algorithm for Dogs and Cats



they can be ventilated using a tight fitting mask. **Supraglottic airway devices can also be used in place of intubation in rabbits and hedgehogs** (the rabbit smallest size V-gel works in hedgehogs). Likewise, intravenous catheters may be very difficult to place in an hypotensive small mammal. An **intraosseous catheter is an adequate alternative** and can be placed in either the tibia or femur. Please refer to your analgesia and anesthesia lecture for more details on this.

For advanced life support, monitoring should be initiated with ECG and capnography (ETCO2). The goal of CPR is to produce at least 30% of normal cardiac output, so ideally we should be reaching 30% normal ETCO2 with quality CPR. Nasal capnography may be used in obligate nasal breathers when intubation is not possible. It can be challenging to place ECG leads on small patients while performing CPR. Doppler unit or ultrasound may also be used to assess heart beat/contractility. Peripheral pulse oximetry becomes nonfunctional during cardiopulmonary arrest and therefore is not recommended for CPR monitoring. Defibrillation can be used for animals weighing more than 500g as it is limited by the lowest settings available on most defibrillators (1-2 Joules). Pediatric internal paddles can be used.

Record keeping during CPR is also important for running an efficient and successful code – see the provided CPR record sheet used by CAPE on the next page.

STABILIZATION

General Plan

The general plan is basically the framework to approach most emergency and critical patients. In order to stabilize patients effectively, you need a minimum amount of information on critical physiological functions that usually maintain body homeostasis. These include **tissue perfusion and oxygenation, electrolyte and acid-base balances, glucose metabolism, and body temperature.** The degree of abnormalities and alterations of body homeostatic function will dictate how aggressive you should be. This critical initial information must be obtained very quickly by doing a brief physical examination (go straight to what is important such as blood pressure, clinical endpoints of perfusion, body temperature, mentation, presence of respiratory distress and so on), and by the judicious use of STAT diagnostic tests (quick tests that directly help to stabilize the patient, but a complete work up must still be done later). Exotic animals decline and die faster and are harder to stabilize, so you need to be quick. Stabilization and information gathering are typically done at the same time. Once the animal is more stable for more comprehensive and/or targeted diagnostic tests, you should do that. You need to get to the diagnosis as soon as you can, because this will dictate the treatment of the underlying process.

In most scenarios, critical animals will be hypotensive or dehydrated and will need some types of fluid therapy with or without vasopressors and fluid additives to correct other metabolic disorders such as acid-base abnormalities, electrolyte abnormalities, and hypoglycemia. If sepsis is suspected, animals should be started on intravenous antibiotics (best if you thought about doing a blood culture first, but it depends on the PCV and size of the animal). Hypoglycemic animals (typically either juveniles, septic patients, or ferrets/guinea pigs with insulinoma) will benefit from dextrose boluses.

STAT Diagnostics

These diagnostic tests are very important for initial assessment of the patient and to guide fluid therapy, emergency procedures, and to help generate a quick working differential diagnostic list and diagnostic and treatment plans. Again, they are done at the same time as stabilization procedures such as placing a catheter and providing oxygen via facemask.

Firstly, they help with the fast detection and quantification of abnormalities that must be immediately corrected. These mostly include hypotension, hypoglycemia, azotemia, acid-base disorders, and electrolyte disorders (ionized calcium, potassium, sodium, chloride).

Secondly, they help with the fast detection of obvious abnormalities that immediately affect emergency treatment. For instance, surgical cases need to be identified early as medical treatment alone will not save these patients. Another example is urinary obstruction where you need to unobstruct the patient.

The STAT diagnostics include the following, they can be done in any order, but typically you will try to identify what is the most life threatening first such as blood pressure.

- **PCV/TS:** this will quickly identify the presence of anemia and hemoconcentration. You could also readily suspect hemorrhage if both are decreased or dehydration if both are increased. **Anemia on a rabbit presented for an acute abdomen should raise the suspicion of a liver lobe torsion.**

- **Blood glucose:** this will obviously identify hypo-

Standard Reporting of CAPE Cardiopulmonary Resuscitation Updated 2023

Patient Sticker	**Date:**	**Location of arrest:**
	Doctor:	**WT:__________kg**
	Mammal ☐ **Avian** ☐ **Reptile** ☐ **Other** ☐	**Time of arrest:** ______________ **Time CPR started:** ______________
Time HR MIN SEC	**Species:**________________	**Monitoring: ETCO2 / ECG / POCUS / BLOODWORK /** ________

HR	MIN	SEC	
			Compressions started ☐ Chest compression type:________________________ Arrest Witnessed (Y/N):
			Ventilation started ☐ ET Tube ☐ Size:__ Mask ☐ LMA ☐ Size:____
			Vascular Access ☐ IVC size:________ Location:___________ Peripheral ☐ Jugular ☐ Central ☐ IO ☐
			CPR Stopped ☐ Reason: ROSC ☐ (Time:_____) Unresponsive to ALS ☐ Owner Decision ☐ Clinician decision ☐

Time HR	MIN	SEC	Monitoring - every 2 min cycle						mmHg	EPI mL	GLYC mL	ATRO mL	_____ mL	_____ mL	_____ mL	Route IV/IT/IO	Palpate Abdomen	Fluids/other drugs/other events
			PEA ☐	Asystole ☐	VF ☐	VT ☐	Brady ☐	ROSC ☐	CO2:									
			PEA ☐	Asystole ☐	VF ☐	VT ☐	Brady ☐	ROSC ☐	CO2:									
			PEA ☐	Asystole ☐	VF ☐	VT ☐	Brady ☐	ROSC ☐	CO2:									
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			PEA ☐	Asystole ☐	VF ☐	VT ☐	Brady ☐	ROSC ☐	CO2:									

Comments:

CPR Team **Number of people:** __________	**Drug Concentrations:**	**Add'l Conc.**	**Fluids Used:**
Lead: Recorder:	Epinephrine	____________:	Crystalloid:
Rotators:	Atropine	____________:	Colloid:
	Glycopyrrolate	____________:	Blood Prod:
ROSC >20min **Yes** ☐ **No** ☐ **Euthanized** ☐	**Debriefed after CPR code** ☐		________:

glycemia, most often caused by **sepsis** (or insulinoma in ferrets/guinea pigs) and hyperglycemia (stress, shock, diabetes). Hypoglycemia needs to be corrected right away as it is life threatening. **Hyperglycemia, especially when severe, has been associated with a poor prognosis and also obstruction in rabbits**. You should never give intravenous dextrose without first taking a glycemia as most rabbits and other small mammals tend to be hyperglycemic when in shock. Also **be aware that glucometers are less reliable in exotic patients and tend to underestimate blood glucose.**

- Urinary specific gravity / urinary dipstick: please go to the lecture on the urinary system for more information. This will typically be performed on a free catch or bottom of the carrier sample. This will help with assessing renal function (low USG on a dehydrated animal is compatible with renal disease) and for the detection of glycosuria and hematuria.

- Indirect blood pressure: IBP is routinely taken using a Doppler unit and a sphygmomanometer. This is less reliable in exotics because of their small size, but can still be valuable. Rabbits are bigger than most other exotics we see and the IBP is more reliable when taken from the frontlimb. Oscillometric are not reliable in most exotic species. It can still be useful to track trends.

- ECG: I would say that it is a lower priority, but if arrhythmias are identified, it is definitely useful. For resuscitation also, so you can quickly see if there is cardiac activity and diagnose ventricular tachycardia or fibrillation.

- Fecal occult blood test is more useful in herbivores as it is less likely to cross-react with food items. This will quickly confirm the presence of melena.

- Venous blood gas and electrolyte panel: this is one of the most important STAT diagnostic test, but unfortunately not always available to private practitioners because of the expense of the blood gas analyzers. Fortunately, they are some portable alternatives that are cheaper (but obviously not as good) such as the Abaxis iStat. **This test is essential to identify blood gas (pH, pCO2, pO2, sO2) and electrolyte (HCO3, Ca, Cl, Na) abnormalities that need immediate correction through fluid therapy, adequate fluid selection, and fluid additives.** Most panels also have additional analytes on their panels such as glucose, lactates, and hemoglobin. There are a few calculated values that you can get too such as the anion gap (AG) and the modified strong ion difference (SID=Na-Cl). **Lactates can be very high in rabbits**, and we don't usually worry until it gets over 15-20 mmol/L. We'll spend a bit more time on those since it is so important. The main things you are trying to identify in small mammals' emergencies are the following:

- Metabolic acidosis: the most common disorder by far. You need the AG or SID to differentiate the 2 main types. **High AG normochloremic (normal SID) metabolic acidosis** is caused by azotemia (hyperphosphatemia actually), high lactates, and diabetes among other causes. **Normal AG hyperchloremic (low SID) metabolic acidosis** is caused by dehydration, renal tubular acidosis, and diarrhea with loss of sodium.

- Metabolic alkalosis: the main type you will see is the **hypochloremic (high SID) metabolic alkalosis, which is almost pathognomonic for a gastric obstruction** (sequestration of HCl in stomach, does not occur with SI obstruction as Cl is reabsorbed in intestines) or vomiting. In rabbits, in particular, this finding should trigger an increased stress level because of the possibility of going to surgery for a gastrotomy.

- Sepsis: this will often be associated with **hypoglycemia and ionized hypocalcemia in our species**. So the combination of those should raise your suspicion for sepsis, which can be diagnosed in patients that meet SIRS criteria and have a confirmed source of infection (e.g. blood culture, intracellular bacteria on cytology).

- Acute kidney injury / obstructive urolithiasis: this will be associated with anuria and resulting hyperkalemia, severe metabolic acidosis, and azotemia.

-Point-of-care (POC) biochemistry: This is different from a full biochemistry panel in that the analyzer is more portable and the results quicker, but the panel is more restricted. The main one I am talking about is the Abaxis Vetscan. This only takes 0.1 mL of whole blood for a biochemistry panel. The panel typically includes at least creatinine, urea, liver enzymes, bilirubin, and electrolytes. So even without a blood gas machine, you can still get some useful information on electrolytic abnormalities. This should be followed up though by a more comprehensive biochemistry panel when the reference lab is open or more blood can be taken from the patient.

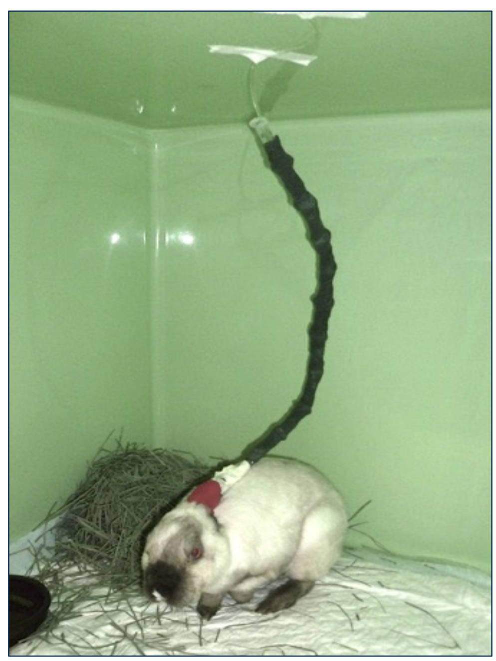
- Point-of-care ultrasound (POCUS): this is super duper useful and may quickly diagnose a wide range of disorders as well as screen and monitor for fluids. For exotic animals, you need a **higher frequency probe** than for domestic carnivores, so about 8-10 mHz.

You don't need to be a diagnostic imaging specialist to do that as you are only looking for obvious things such as the **presence of cavitary fluids** (pericardial, pleural, abdominal) and **specific diseases** such as liver lobe torsion (use color doppler to screen for vascularization of liver lobes), ovarian cysts in guinea pigs, urolithiasis, and obvious masses (use color doppler to look for vascularization and differentiate masses from abscess for instance).

Fluid Therapy - Technical Aspects

Fluid therapy follows the same basic principles as for any animal. The first step is obviously to gain vascular access. Fluid therapy can be given subcutaneously, but there is just better hydration and distribution of fluids with IV administration and you can more finely tune protocols. **Subcutaneous fluid therapy should only be used for mild disease.**

The **cephalic and saphenous veins** are accessible in most species. In rabbits, it is harder to give fluid chronically through a saphenous vein catheter as the vein frequently occludes when the leg is flexed, so cephalic is preferred. In rabbits again, you can also use the **marginal ear vein.** It is actually quite handy as the fluid line is not in front of the rabbit and can be attached to the top of the enclosure. Because of the added weight on the ear, ears should be tapped together to prevent cartilage issues.



The big drawback of ear catheters is that they are associated with ear necrosis, which can be minimal to very extensive. When using an ear catheter, the owner should be aware of that risk (sometimes, it is the price to pay to have them survive), you should not have bandages too tight, and you should **NEVER inject drugs that are not tissue friendly in an ear catheter**. In fact, you should try to stick to just fluids and vasopressors and if you inject any drugs, make sure you flush very well after. In case you are wondering, the ear necrosis is associated with thrombosis of arterioles because of the extensive arteriovenous anastomosis in the ears of rabbits.

All IV lines should be protected from biting. Rabbits, rodents, and ferrets will bite these lines. As a matter of fact, it may be impossible to keep an IV line in a healthy ferret for instance.

When you cannot get IV access because the animal is too small and hypotensive, the plan B is to get an **intraosseous catheter.** This is mainly done in rodents or youngsters as we often get an IV line one way or another into adult rabbits and ferrets. In rodents, the **femur or tibia are typically used f**or this purpose.

Fluid Therapy - The Fluid Therapy Plan

There are 3 phases to your typical fluid therapy plan:

- Phase I: resuscitation phase. That's when you are basically treating shock. **Most exotic mammals present in decompensated shock.** Hypovolemic and septic shocks are quite common. You are the most aggressive during this phase with high fluid rate, fluid boluses, and vasopressors. We usually start by giving a few **10-15 mL/kg crystalloid fluid boluses IV** while we ready fluid lines, vasopressors, and fluid pumps. For severe shock, boluses of **3-4 mL/kg of hypertonic saline** will be more effective at quickly expanding vascular volume. **We then quickly add vasopressors to increase blood pressure and tissue perfusion.** Also, septic shocks and some shocks we see in exotic mammals tend to be poorly fluid-responsive and our species are prone to fluid overload (which will typically result in pleural effusion). **The vasopressor of choice is norepinephrine. Dopamine can also be used, but it does not work in rabbits and is poorly effective for sepsis.** Norepinephrine rate in ferrets is similar to cats, but it is much higher in the rabbit at 0.5 to 2 ug/kg/min. Doing a quick blood gas may also help to correct some severe abnormalities at the same time (making sure you don't exceed certain rates such as for Na and K correction). As you resuscitate your patient, you should evaluate the response: improvement in clinical endpoints such as mentation, BP, CRT, HR, pulse, and blood gases values.

-Phase II: rehydration phase. If you got out of phase I, congratulations, you earned the right to rehydrate your patient. So, that's when you calculate a good **fluid rate that will depend on the daily maintenance for that particular species (higher in rabbits and rodents at 100 ml/kg/day), the dehydration percentage, and ongoing losses (diarrhea, vomiting, polyuria).**

Percent dehydration	Clinical signs
<5	No detectable abnormalities
5-8	Decreased skin turgor, dry mm
8-10	Decreased skin turgor, dry mm, eyes sunken, slightly prolonged CRT
10-12	Severe skin tenting, prolonged CRT, dry mm, eyes sunken, possible signs of shock
>12	Above, signs of shock, life threatening

The plan is to correct fluid deficits over 12 to 24h. The more acute the dehydration occurs, the quicker you could correct these deficits.

Because exotic mammals are small, we tend to use more **syringe pumps** than fluid pumps in them. In terms of fluid selection, a balanced electrolytic solution will do well in most cases. For hypochloremic metabolic alkalosis and hypochloremia, you need to either supply additional chlorides or an acidotic fluid and 0.9% NaCl is indicated in these situations. There is a range of **fluid additives** you can use to correct particular abnormalities. Most fluids are deficient in potassium so you will need to supplement with potassium for more chronic fluid therapy (do not exceed 0.5 meq/kg/h). Bicarbonates ($NaHCO3$) are used in certain prolonged metabolic acidosis or after CPR. Calcium gluconate is used to correct ionized hypocalcemia. Insulin/dextrose combination is used for severe hyperkalemia (intracellular translocation of K). Dextrose 50% is used to supplement fluids at 2.5 to 5% to treat ongoing hypoglycemia.

You can also use **CRI** of some useful drugs such as more potent pain meds such as fentanyl, continuing on using norepinephrine, and lidocaine as a GI promotility drug and analgesic in gastric stasis in rabbits. It is important to take into account the total volume of fluids being administered across all CRIs and adjust fluid rate accordingly.

-Phase III: maintenance phase. After all we went through, the maintenance phase is a piece of cake. That's basically when you only provide maintenance needs, so about 50-100 ml/kg/day depending on the species. In small mammals, this phase is typically performed using subcutaneous administration. Rabbits and rodents have fairly loose skin and can accomodate an impressive amount of SC fluids.

Ferrets, on the other side, hate SC fluids and it can be hard to give to some of them.

-Blood transfusion: you may need a blood transfusion if the animal is severely anemic or actively bleeding. The indication is **based on clinical signs in combination with a PCV lower than 15-20%, especially with acute loss.** The most common indication in our service is for rabbits with liver lobe torsion. You will need a donor as there is not small mammal blood bank. The donors need to be healthy, young, and big. Fortunately, there is **no blood groups in ferrets,** so you can do whatever you want. In rabbits, it does not seem to matter as well, you just need to do a **cross-match** before. The expectations is to raise the PCV by 1% for every 2 ml/kg given. You need to use a neonatal **blood filter** so there is no clots transfused. The administration rate should initially be low for 20 minutes to monitor for transfusion reaction. Then, it can be increased in order to deliver the whole transfusion within 3-4h. Whole blood is considered as a colloid and you should be careful not to fluid overload with the other line of crystalloid.

Heat Support

Heat support is essential and can initially be overlooked when doing CPR and initial resuscitation. However, our **animals lose heat extremely rapidly,** not only they present hypothermic when in shock, but further lose a lot of heat with the added alcohol being poured over them for catheters, blood sampling, monitoring equipment placement, and POCUS. **Severely hypothermic animals will not regain normal heart rate, blood pressure, and mentation until they are warmed up.** On the plus side, our animals can get rewarmed fast because of their high surface-to-volume ratio. There are different ways of rewarming animals. We typically use either **condutive sources such as heating pads** (the Hotdog™ is particularly effective, oat bags are effective and cheap) or **convective sources such as forced warm air** (Bairhugger™ is the main one used). Fluids used in fluid therapy should be prewarmed at body temperature. Once animals are more stable, they should be placed in **warm incubators** until they are normothermic.

Oxygen Therapy

Oxygen therapy is indicated for any dyspneic patient, but is not equally effective for all cases of respiratory distress. It tends to be less effective for instance for mechanical respiratory diseases (upper respiratory diseases) than for parenchymal pulmonary diseases or diseases with high ventilation-perfusion mismatching. In any case, we just give it to any dyspneic patient whether they are hypoventilating or hypoxemic.

The goals of oxygen therapy are to deliver a high fraction of inspirated of oxygen (FiO2) to improve blood oxygenation, hemoglobin saturation, and tissue delivery of oxygen.

Small mammals have a low tolerance for invasive means of providing oxygen therapy such as nasal cannula. Therefore, we use more hands-off techniques such as **flow by (FiO2=25-40%), face masks (FiO2=35-60%), and oxygen cages (30-60%)**. Make sure you double check the FiO2 your oxygen cage is able to deliver using an oximeter.

Nutritional Support

Many patients presented on emergency are anorexic or in a state of negative energy balance. Because of their high metabolism, they also further lose weight quickly. Once stabilized, nutritional support should be initiated providing there are no contra-indications. There are multiple ways of providing nutritional support. Non-invasive methods include mainly **syringe and assisted feeding**. If oral administration is contra-indicated or animals are very stressed by repeated hand-feeding of large volumes, more invasive techniques should be considered. In specific cases of advanced maxillofacial diseases, these techniques may also be used. **In rabbits, and guinea pigs, you can place nasogastric tubes** and feed a more liquid diet. Since they are obligate nasal breathers, you need to keep an eye on their breathing and rhinitis is a contra-indication. In ferrets, gastrotomy tubes can be placed.

The diets that are fed are **higher calorie critical care formula made for either herbivores or carnivores**. Two commonly used brands are Oxbow and EmerAid. Rabbits and hystricomorph rodents need to be fed about 50 ml/kg/day divided into 2-4 feedings using either a catheter tip 60cc syringe or 1cc syringes repeatedly, going into the interdental space.

COMMON EMERGENCIES

Emergency presentations are obviously very varied. With that said, there are some that are more common than other or that are just more urgent. Respiratory emergencies, gastrointestinal emergencies, trauma, urinary obstruction, and acute abdomen particularly stand out. Please refer to the corresponding lectures for more information on these respective organ systems. Below is a quick table of probably the most common emergency presentations by species.

Rabbits	Rodents	Ferrets
Urinary obstruction (urolithiasis)	Urinary obstruction (urolithiasis)	Urinary obstruction (urolithiasis, prostatomegaly)
Respiratory distress	Respiratory distress	Respiratory distress, cardiac
Diarrhea, enterotoxemia	Diarrhea, enterotoxemia	Diarrhea
GI stasis syndrome	GI stasis syndrome	GI foreign body
Trauma (fall, predators, back injuries)	Trauma (fall, predator)	Trauma (fall)
Liver lobe torsion	GDV (guinea pigs)	Insulinoma "crisis"
Head tilt/neurological signs		

We will spend more time on a few important topics, but will only highlight the emergency management side of these disorders.

The Dyspneic Rabbit

Signs of respiratory distress in rabbits include tachypnea, increased respiratory efforts, nasal flaring, assuming an orthopneic position (nect extended), open mouth breathing (very concerning), and intolerance to effort.

There are many potential respiratory diseases of the rabbit, but you should at least have the following as part of your differential diagnosis depending on the age and other risk factors:

-Pneumonia

-*Pasteurella multocida* infection

-Rhinitis (often odontogenic)

-Thymoma (it presses on the trachea)

-Respiratory neoplasia (e.g. pulmonary metastasis of uterine adenocarcinoma)

-Cardiac causes and pleural effusion

Below is a summary of management considerations for this type of presentation.

Stabilization	STAT Diagnostics	Full Work-up	Treatment
• Cleaning nostrils • Midazolam • Oxygen therapy • Have v-gel/ ET tube ready to go	• POCUS: T-fast, A-fast • Blood gas / electrolytes	• CBC/chem • Radiographs • Thoracic US/ echocardiography • Skull/thorax CT • Rhinoscopy • Cytology/culture (deep nasal swab, FNA of consolidated lung lobes or masses)	• Antimicrobial • Nebulization • Thymoma: radiation therapy, prednisolone • Rhinitis: rhinotomy

The Rabbit with GI Syndrome (Stasis)

There is a tremendous variation in how these cases present. **Gastrointestinal stasis is not a disease in itself, it is a syndrome**, so presentation, outcome, and treatment will also heavily depend on the underlying disease process. With GI stasis, rabbits typically present in pain (bruxism, increased facial grimace score, lack of activity), **anorexia, with a lack of stool production**, dehydrated, and with an enlarged stomach and stometimes bloating.

The differential diagnosis includes **husbandry-related risk factors such as a lack of non-digestible fibers in the diet, increased carbohydrate consumption, hair ingestion, stress, sources of pain; dysbiosis, enterisis, enterotoxemia; dental disease; obstructive conditions such as caused by food and hair (trichobezoar); cecal diseases such as typhlitis, appendicitis, and cecal impaction; and non GI-related diseases causing anorexia and/or visceral pain such as liver lobe torsion, urolithiasis, encephalitozoonosis, uterine adenocarcinoma, and otitis media.**

Below is a summary of management considerations for this type of presentation.

Stabilization	STAT Diagnostics	Full Work up	Treatment
• Fluid therapy (SC/IV) • Obstruction: stomach decompression with orogastric tube • Vasopressors: norepinephrine • Heat support • Analgesia: fentanyl, buprenorphine • Sepsis: IV antibiotics	• Blood pressure (Doppler) • Blood gas / electrolytes • POCUS: A-fast • POC biochemistry: glucose, lactates, liver enzymes • USG	• CBC/chem • Radiographs • Abdominal ultrasound • CT • Fecal flotation • Urinalysis	• Obstruction: enterotomy, gastrotomy • Fluid therapy • Antibiotics: TMS, metronidazole, fluoroquinolone • Analgesics (opioids, meloxicam) • GI prokinetics: lidocaine, maropitant • Syringe feeding (oral, nasogastric tube) • Physical exercise

The Liver Lobe Torsion Rabbit

This should be suspected in **any rabbit presented for an acute abdomen** or any acute presentation really. These cases are **typically anemic, azotemic, and have elevated liver enzymes (AST, ALT)** on biochemistry and there are not a lot of diseases in rabbits with this presentation, so it should really be in your differential at that point. An **A-FAST focusing on the caudate lobe** (deep on the right, near the kidney) should be performed. When using color doppler, you may see a lack of vascularization in this lobe compared to other lobes. You may also notice some abdominal fluid (which is blood in these cases, that's why they are anemic).

Stabilization is as described above. These cases are very painful and can be very unstable. Initiate fluid therapy, analgesics, and vasopressors right away. Doing a **blood transfusion** or preparing one for perioperative administration is also recommended. The treatment of choice is lobectomy of the affected lobe. However, if funds are tight for the owner, some of these rabbits actually make it on medical management only and there are many cases of incidental liver lobe torsion being found on necropsy. In a recent study, no significant difference was found between the seven day survival rates for medical versus surgical management, however rabbits that underwent liver lobectomy had three times longer mean survival time compared to medical management. Whether rabbits should go to surgery also depends on the severity of clinical signs. Patients with more severe presentations are more likely to require lobectomy and surgical hemostasis to survive.

GI Emergencies in Ferrets

Gastrointestinal emergencies are probably the most common type of emergencies seen in pet ferrets. They present with signs like diarrhea (probably the most common GI sign in ferrets), melena, anorexia, lethargy, vomiting (less common than in other carnivores), and abdominal pain.

The differential diagnosis can be extensive for this in ferrets. The signalment and history may make some diseases more likely than others. **You should try to rule out foreign bodies first** (palpation, radiographs, ultrasound) as those are surgical cases and ferrets may only show diarrhea with those. These differentials include the following:

-Gastric ulcers caused by toxins (ibuprofen), *Helicobacter* gastritis among many causes.

-GI foreign bodies.

-Epizootic catarrhal enteritis caused by the ferret coronavirus.

-Inflammatory bowel disease.

-Eosinophilic gastroenteritis.

-Proliferative bowel disease caused by *Lawsonia intracellularis*.

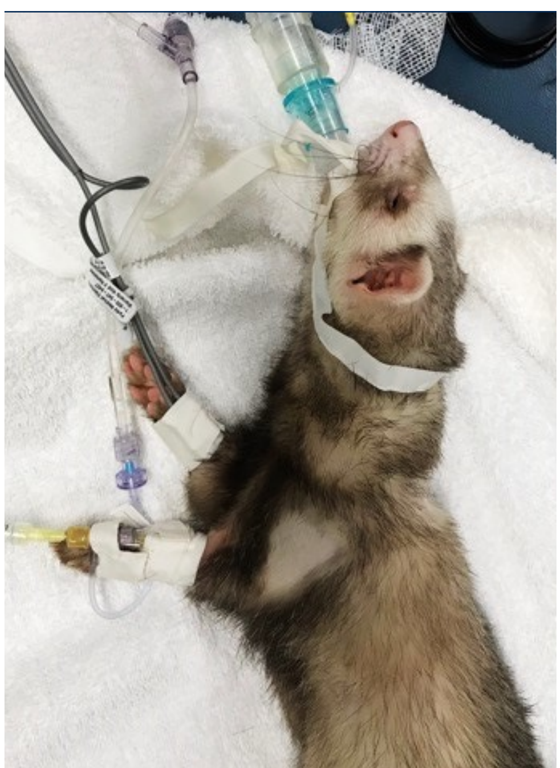
-Enteritis caused by *Salmonella*, *Campylobacter* or other bacteria.

-GI lymphoma.

-GI parasites.

Below is a summary of management considerations for this type of presentation.

Stabilization	STAT Diagnostic	Full Work up	Treatment
-Fluid therapy (IV) -Vasopressors: norepinephrine, dopamine -Heat support -Analgesia: fentanyl, buprenorphine -Sepsis: IV antibiotics	-Blood pressure (Doppler) -Blood gas / electrolytes -POC biochemistry -USG -POCUS: A-fast	-CBC/Biochem -Whole-body xrays -Abdominal ultrasound -Fecal flotation -GI biopsy -PCR for infectious agents	-FB: enterotomy, gastrotomy -Fluid therapy -Antibiotics: TMS, metronidazole, triple therapy for helicobacter -Analgesics (opioids, meloxicam) -Gastric protectants: omeprazole, famotidine -Antiemetic: maropitant -Assisted feeding (critical care diet) -IBD: immunosuppressive meds



The Ten Most Important Facts

1. Exotic mammal species tend to hide clinical signs and consequently will present in a state of decompensation on emergency. Initial triaging should focus on obtaining a brief history targetting signalment and the presenting issue and performing a brief physical examination including mentation, signs of volemia/ perfusion, signs of dehydration, core body temperature, quick abdominal palpation, and cardiopulmonary assessment.
2. CPR are similar in small mammals than in other species. The main difference concerns airway access in rabbits and rodents. Tight-fitting masks or laryngeal masks can be used for this purpose. Intravenous catheters may also be challenging to place and you should consider intraosseous catheters in some intances.
3. Initial stabilization involves correcting all identified homeostatic abnormalities. These are frequently identified by a combination of physical examination and performing STAT diagnostic tests. Once stable, a more comprehensive diagnostic work-up should be obtained to identify the underlying disease process.
4. STAT diagnostics include, at minimum, PCV/TS, blood glucose, and indirect blood pressure measurement. If available, a blood gas / electrolyte panel and a point of care ultrasound should be performed as they yield a huge amount of information in a short timespan.
5. Metabolic acidosis is the most common type of acid-base abnormalities associated with azotemia, dehydration, and diarrhea among other causes. Metabolic alkalosis is encountered in gastric obstruction and vomiting. Sepsis is characterized by hypoglycemia and ionized hypocalcemia on an emergency panel.
6. POCUS is used to screen for cavitary fluids and screen for common diseases in easy-to-image approaches such as targetting the caudate liver lobe of the rabbit, looking for large calculi in the bladder, or searching for an obvious abdominal mass.
7. The fluid therapy plan is divided into 3 phases: the rescussitation phase which involves fluid boluses, high rates of fluids, and vasopressors; the rehydration phase which focuses on correcting fluid deficits; and the maintenance phase which is typically administered subcutaneously in small mammals.
8. Heat support is important in exotic mammals as they lose heat rapidly and frequently present hypothermic. Conductive (heating pads) and convective (forced warm air) sources are most often used. Fluids should be prewarmed before administration.
9. Oxygen therapy aims to increase the fraction of inspired oxygen to promote tissue delivery of oxygen. It is administered by flow-by oxygen, facemaks, or in an oxygen cage.
10. Nutritional support is provided in the stabilized patient. Many exotic mammals are anorexic when sick and should be provided with syringe feeding of high calorie liquid critical care diets for herbivores or carnivores.